

Observing users

Chapter 12

Observation

- Why? Get information on..
 - Context, technology, interaction
- Where?
 - Controlled environments
 - In the field (where the product is used)
- Observer:
 - outsider
 - participant
 - ethnographers

Frameworks to guide observation

- - *The person*. Who?
 - *The place*. Where?
 - *The thing*. What?
- The Goetz and LeCompte (1984) framework:
 - *Who* is present?
 - What is their role?
 - *What* is happening?
 - *When* does the activity occur?
 - *Where* is it happening?
 - *Why* is it happening?
 - *How* is the activity organized?
- Checklist can also help (p. 369).

Data collection

- Notes:
 - not technical, writing speed may be a factor, hard to observe and write at the same time, laptop is faster but intrusive and cumbersome, two people work better than one.
- Still camera:
 - images are easily collected, allows evaluators to be mobile.

Data collection cont.

- Audio:
 - less intrusive than video, allows evaluators to be mobile, inexpensive, lack of visual records, hard to transcribe data.
- Video:
 - both visual and audio data, can be intrusive, can be inexpensive with small cameras, can allow evaluators to be mobile, attention is focused on what is seen through the lens, analysis can be time consuming.

Data collection cont.

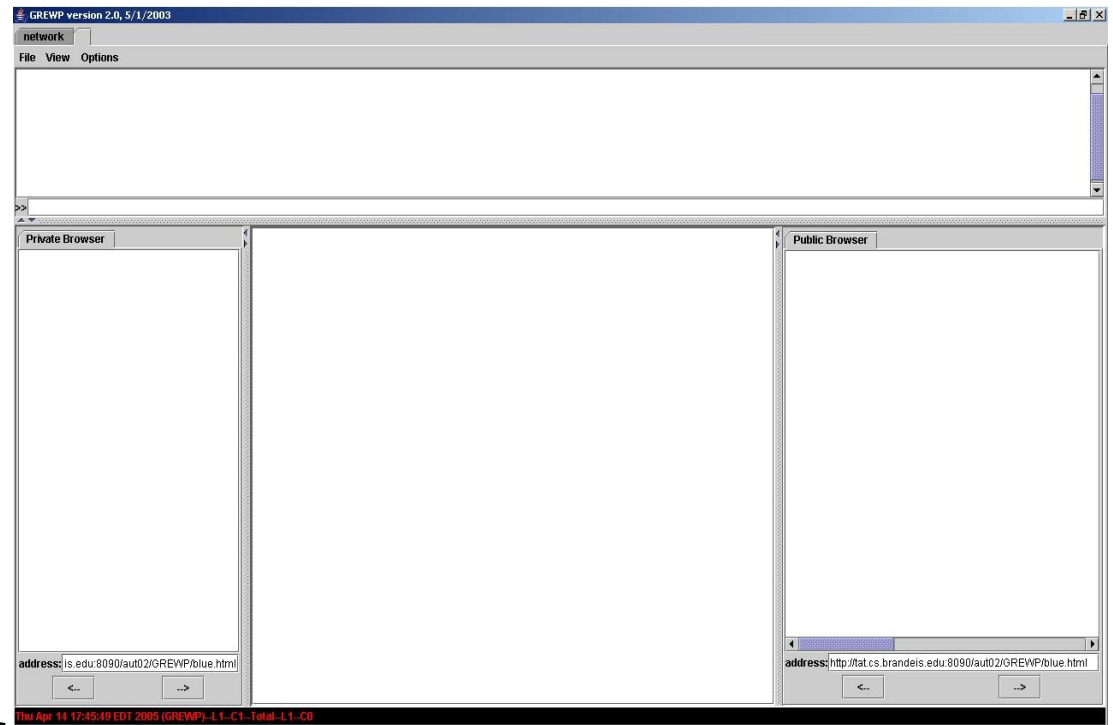
- Interaction logging (transcripts & replay):
 - logs everything you do in the system, easy to generate detailed analysis, transparent to the user, facial expression etc. is not logged.
- Techniques may be used individually or combined => requires coordination.

Data analysis

- *Qualitative data - interpreted* & used to tell the ‘story’ about what was observed.
- *Qualitative data - categorized* using techniques such as content analysis.
- *Quantitative data* - collected from interaction & video logs. Presented as values, tables, charts, graphs and treated statistically.

CS111 Experiment

- Create a presentation of a world country and its culture.
- ***GREWP tool provides users with:***
 - *a shared workspace online,*
 - *chat to communicate,*
 - *public and private browsers*
 - *Generates transcripts for replay*



Observing the users

- Where was the study performed:
 - Controlled environment
 - We had everything set up before participants arrived
 - Tested the software
 - etc.
- Data collection:
 - Note taking: Important issues noted on paper and coordinated with transcripts later.
 - Replayed transcripts that the tool generated.

Analyzing the data

- Coordinated notes with transcripts
- Replayed the transcripts
 - Qualitative data (Categorization)
 - Looking for incidents or patterns.
 - How was a certain task completed?
 - How did the users use a certain component in the system?
 - One user frequently got stuck in the HTML coding. Why is that?
 - Analyzing the discourse (Alex Feinman)

network vcr

File Font Options Tabs

Replay Events

user2: or some african?
 user2: what's about south africa?
 user1: I just thought the same
 user1: Here we go
 user2: which topic about south africa?
 user1: perhaps we start with some facts about history, but I think we have to short them dramatically

Private Browser

ERROR: java.io.IOException

```

An Introduction to
</title>
</head>
<body bgcolor="silver">
<h1>Headline h1 for the webpage</h1>
<h2>Headline h2 for the webpage</h2>
<h3>Headline h3 for the webpage</h3>

<table border="2" align="center" bgcolor = "white">
<tr>
<td>
Text format examples:
<br>
<b>Bold text</b>
<br>
<i>Italic text</i>
</td>
<td>
<font size="2" face="Verdana" color="red">

```

- [A History of South Africa, Third Edition](#) By Leon [www.amazon.com](#)
- [Research South African History](#) Find quality info at the [www.questia.com](#)

WEB RESULTS

1.

[South African History South Africa History](#)

South African history. ... Cable and Wireless, a History - South Africa 1920's to

[www-sul.stanford.edu/depts/ssrg/wbr](#)

2.

Thu Apr 14 17:36:09 EDT 2005 (GREWP)--L4--C20--Total--L57--C20

PANIC

watch mode ▾

eval-html

validate

Name Value

admin VIEW	editor
admin CH...	3
user1 CHAT	7
user2 VIEW	chat
user2 CHAT	8
user1 VIEW	chat
admin CO...	21
user1 CO...	6
user2 CO...	20

LogFile:

3\Computational Cognitive Science CS111\cosi111\Experiment\Experiment\Session 1\log-admin-user1==user110022003-1065

time

Thu Oct 02 14:38:37 EDT 2003

event

(1065119917250L (GroupFWChat groupchat)

status

play

stop @ events: ☐ chat ☐ watch/edit ☐ form ☐ eval ☐ panic ☐ window-switch

event number lastevent elapsed time total time time to next event

229	7452	3119.438	9815.906	0.0
-----	------	----------	----------	-----

-----FOLLOW ALL----- ▾

<

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-

Redesign

user2: look where I'm in the screen
user2: title is only in the head
user2: not in the normal text

user2: look how I to a table
user2: you only put title only in the head

- Proposal 1:
- *Automatically add reference to a line in the code window to the chat.*
- *Help users stay coordinated*

Redesign

user2: how is the work?

user2: how far is your table??

user1: where are you now?

user1: are you finished with the food?

- Proposal 2:
- **Provide a way to write down a plan and review or modify it visually.**
- **Helps users be aware of each others work.**
- **Automatic update of the plan as work progresses.**

UNIPLEX is the official **Education Management System (EMS)** used by the **Military Institute of Science and Technology (MIST)** to manage academic activities such as course registration, result publication, attendance tracking, and communication between students, faculty, and administration.

Recently, concerns have been raised by both students and faculty regarding the usability and user experience of the system. Users report difficulties in navigating key features, system delays, and a lack of intuitive feedback during task execution (e.g., course dropping, downloading grade reports). Additionally, some components (such as result views and advisor approvals) require frequent clarification from support staff, indicating possible design or interaction inefficiencies.

To address these challenges, the UNIPLEX development team decided to conduct a structured **user observation study**, following industry-standard approaches. The objective is to evaluate how users actually interact with the system in both **controlled environments** (lab-based usability testing) and **real-world usage** (in the field), using methods and frameworks drawn from **qualitative and quantitative research**.

As part of this initiative, you are tasked with solving the following problem:

How can structured user observation and methods like interaction logging, note-taking, audio recording, and ethnographic observation be effectively applied to uncover usability challenges in the UNIPLEX system at MIST?

Your analysis should focus on:

- Selecting appropriate observation environments and techniques.
- Defining a structured observation framework.
- Collecting and analyzing qualitative and quantitative user data.
- Identifying key usability issues.
- Recommending evidence-based improvements to the system's design and interaction model.

What is UNIPLEX (MIST EMS)?

UNIPLEX at MIST is an integrated **Education Management System** used for tasks like:

- Course registration
- Attendance tracking
- Result publishing
- Assignment submission
- Faculty-student communication
- Class routine management

How Task Analysis Applies to UNIPLEX ???

1. Task Decomposition (HTA) in UNIPLEX
2. Knowledge-Based Task Analysis
3. Entity/Object-Based Analysis

Task Decomposition (HTA) in UNIPLEX

Task: Register for courses

Hierarchical Task Analysis (HTA):

Main Goal: Register for the semester

Subtasks:

1. Log in to UNIPLEX
2. Navigate to course registration panel
3. Select courses as per credit requirement
4. Confirm and submit
5. Download or verify course list

Plan: Do 1 → 2 → 3 → 4 → 5 in that order.

This breakdown helps system designers ensure the interface supports the task sequence logically and clearly.

Knowledge-Based Task Analysis

Task: Register for courses

To successfully register or use UNIPLEX, a user (student) needs to **know**:

- Their credit requirements
- Course prerequisites
- How to navigate the UNIPLEX dashboard
- Course codes and titles
- Deadline for registration

Designing the system should consider this knowledge – e.g., providing tooltips, course descriptions, credit summaries.

Entity/Object-Based Analysis

Task: Register for courses

In UNIPLEX, key **entities (objects)** and **actions** include:

This helps map the system's data models and understand how users interact with features.

Entity/Object

Student profile

Courses

Assignments

Results

Attendance

Associated Actions

View, update, submit

Browse, select, drop

Upload, view feedback

View, download, contest

Check, notify instructor

Classroom Example for Discussion

Example Task: Submitting an Assignment via UNIPLEX

Asking:

- What steps do you follow?
- What knowledge is required?
- What objects and actions are involved?

Expected Answers:

- Steps:** Login → Go to course → Find assignment → Upload → Confirm
- Knowledge:** Deadline, file format, assignment title
- Objects/Actions:** Assignment (upload), Course (navigate), Notification (receive)

Then, let students **draw an HTA tree** or flow diagram of this process.

Summary Table

Task Analysis Method

Task Decomposition

Knowledge-Based

Entity/Object-Based

UNIPLEX Example

Registering courses, uploading assignments

Knowing course codes, deadlines

Student, course, assignment, result

Evaluation Techniques

Evaluation Techniques

- Evaluation
 - tests usability and functionality of system
 - occurs in laboratory, field and/or in collaboration with users
 - evaluates both design and implementation
 - should be considered at all stages in the design life cycle

Goals of Evaluation

- assess extent of system functionality
- assess effect of interface on user
- identify specific problems

Evaluating Designs

Cognitive Walkthrough

Heuristic Evaluation

Review-based evaluation

Task- Centered Design

- Task-centered design focuses on REAL, COMPLETE, REPRESENTATIVE tasks.
- Traditional requirements analysis looks at ABSTRACT, PARTIAL task elements.
- How to evaluate a design without users? One approach is a *cognitive walkthrough*.

Cognitive Walkthrough

Proposed by Polson *et al.*

- evaluates design on how well it supports user in learning task
- usually performed by expert
- Identify problems based on psychological principles
- Requires some expertise in cognitive psychology
- forms used to guide analysis

Polson P., Lewis, C., Rieman, J., and Wharton, C. (1992). Cognitive walkthroughs: A method for theory-based evaluation of user interfaces

Cognitive Walkthrough (ctd)

- For each task walkthrough considers
 - what impact will interaction have on user?
 - what cognitive processes are required?
 - what learning problems may occur?
- Analysis focuses on goals and knowledge: does the design lead the user to generate the correct goals?

Cognitive walkthrough

- Formalized way of imagining people's thoughts and actions when they use an interface for the first time.
- First select a task that the design is intended to support.
- Then try to tell a believable story about each action a user has to take to do the task.
- To make the story believable, you have to motivate each of the user's actions, relying on the user's general knowledge and on the prompts and feedback provided by the interface. If you can't tell a believable story about an action, then you've located a problem with the interface

Cognitive walkthrough

1. Select tasks
2. Write-down the desired path
3. 4 questions to ask:
 - Will the customer realistically be trying to do this action? (Is effect of current action same as user's goal? – conceptual model)
 - Is the control for the action visible?
 - Is there a strong link between the control and the action? (Will user recognize action as the correct one?)
 - Is feedback appropriate? (Will user understand feedback?)

Cognitive walkthrough

- ✓ No need for users
 - ✓ Explicit account of user's task
 - ✓ Suggestions to improve learnability
 - ✓ Any phase
 - ✓ Quick and inexpensive
-
- Expertise skills / evaluator effect
 - Tends to focus on words and graphics
 - No frequency and severity of problem
 - Artificial context

Heuristic Evaluation

- Proposed by Nielsen and Molich.
- usability criteria (heuristics) are identified
- design examined by experts to see if these are violated
- Example heuristics
 - system behaviour is predictable
 - system behaviour is consistent
 - feedback is provided
- Heuristic evaluation 'debugs' design.

Heuristic Evaluation

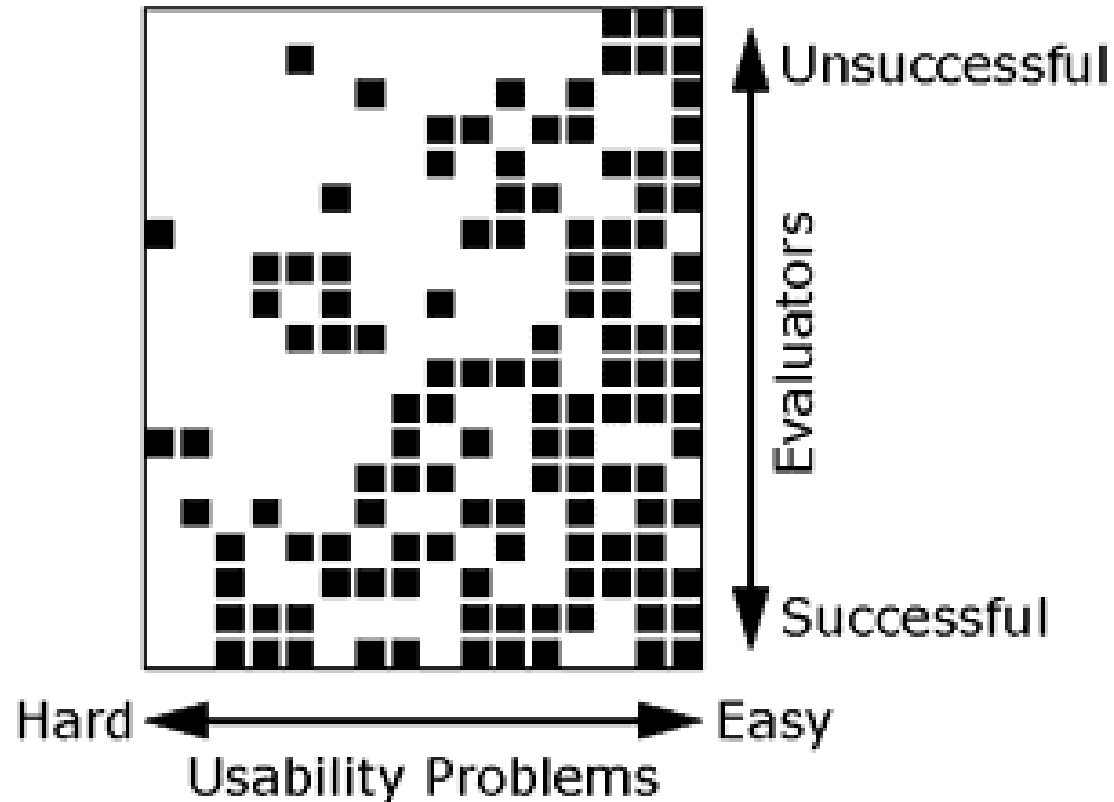
- Several (3 to 5) evaluators independently critique a system to come up with potential usability problems.
- Relatively cheap approach--- *Discount usability* technique
- A set of 10 heuristics are provided to aid the evaluators in discovering usability problems.
- Heuristics are almost same like the *principles* and *guidelines*.

Heuristic evaluation

- Developed by Jakob Nielsen
- Heuristics → general principles for interaction design
 - Adjusted versions for particular cases
- Identify where design does not follow these principles
- Requires some expertise in usability (3-5 evaluators)
 - independently check for compliance with usability principles (“heuristics”)
 - different evaluators will find different problems
 - evaluators only communicate afterwards
 - findings are then aggregated

Why Multiple Evaluators?

- Every evaluator doesn't find every problem
- Good evaluators find both easy & hard ones



Heuristic evaluation process

1. Select system and tasks to review
2. Select heuristic set
3. Inspect the interface alone
4. Communicate and aggregate findings
5. Categorize and prioritize findings to be effectively reported

Severity Rating of HE

- The severity of a usability problem is a combination of three factors:
 - The **frequency** with which the problem occurs: Is it common or rare?
 - The **impact** of the problem if it occurs: Will it be easy or difficult for the users to overcome?
 - The **persistence** of the problem: Is it a one-time problem that users can overcome once they know about it or will users repeatedly be bothered by the problem?

Severity Rating of HE

- The following 0 to 4 rating scale can be used to rate the severity of usability problems:

0 = I don't agree that this is a usability problem at all

1 = Cosmetic problem only: need not be fixed unless extra time is available on project

2 = Minor usability problem: fixing this should be given low priority

3 = Major usability problem: important to fix, so should be given high priority

4 = Usability catastrophe: imperative to fix this before product can be released

Heuristic evaluation

- ✓ Relatively inexpensive (incl. planning)
 - ✓ Expert skills are not a must
 - ✓ Can be used early, but more efficient later
-
- Evaluator effect (thus, multiple are needed)
 - Not as effective as usability testing (minor issues)
 - Not always readily suggest solutions
 - Prone to false alarms

Nielsen's 10 Heuristics

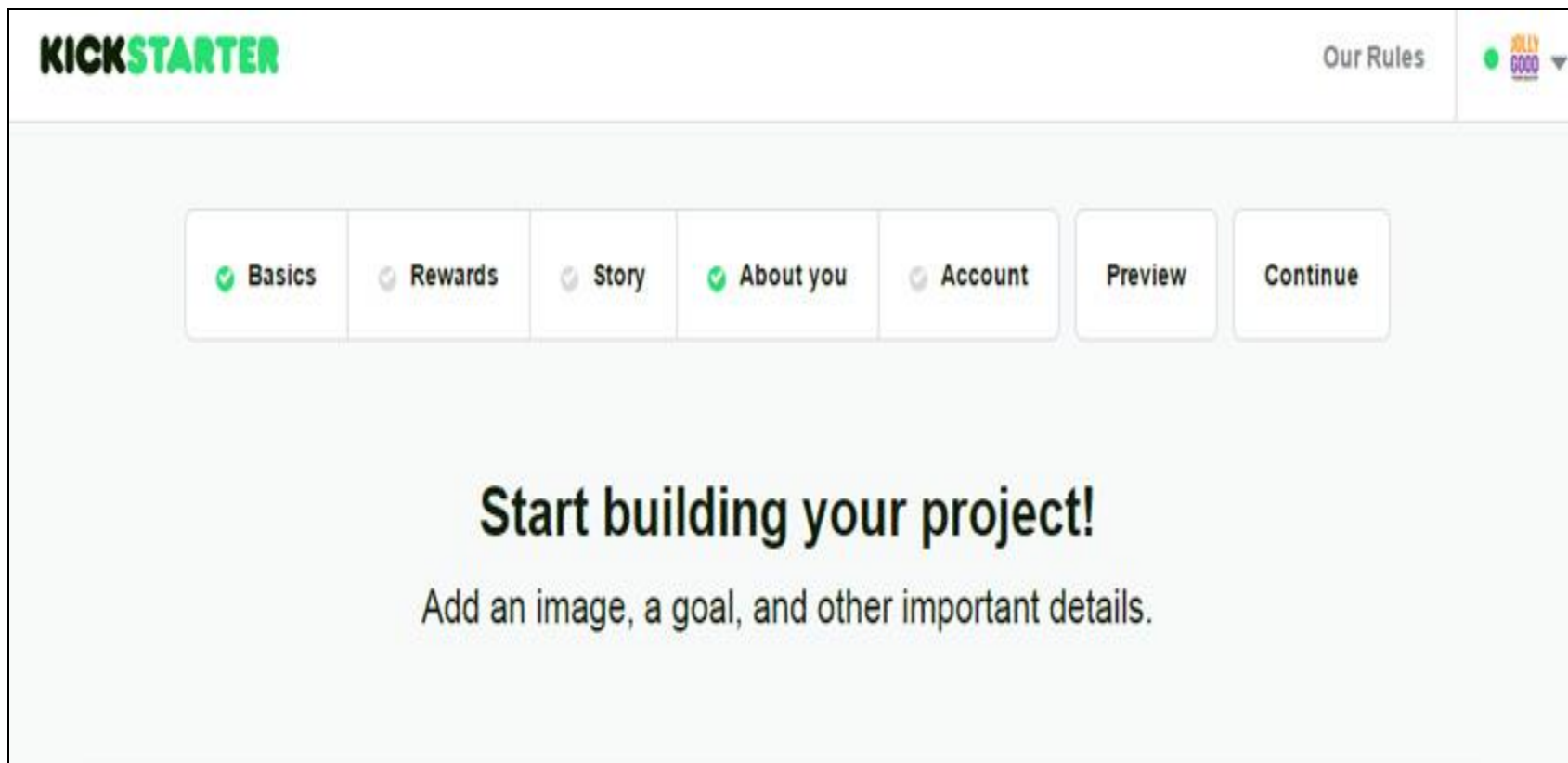
1. Visibility of system status
2. Match between system and the real world
3. User control and freedom
4. Consistency and standards
5. Error prevention
6. Recognition rather than recall
7. Flexibility and efficiency of use
8. Aesthetic and minimalist design
9. Help users recognize, diagnose, and recover from errors
10. Help and documentation

1. Visibility of system status

- Provide users timely and appropriate feedback about the system's status.
- Does the system tell the users
 - what operation it is performing?
 - how much longer they need to wait?
 - which mode it is in?

Example: If it takes a long time to load a screen, display a progress bar and/or an estimate of the time it may take to load, so users know what to expect.

1. Visibility of system status



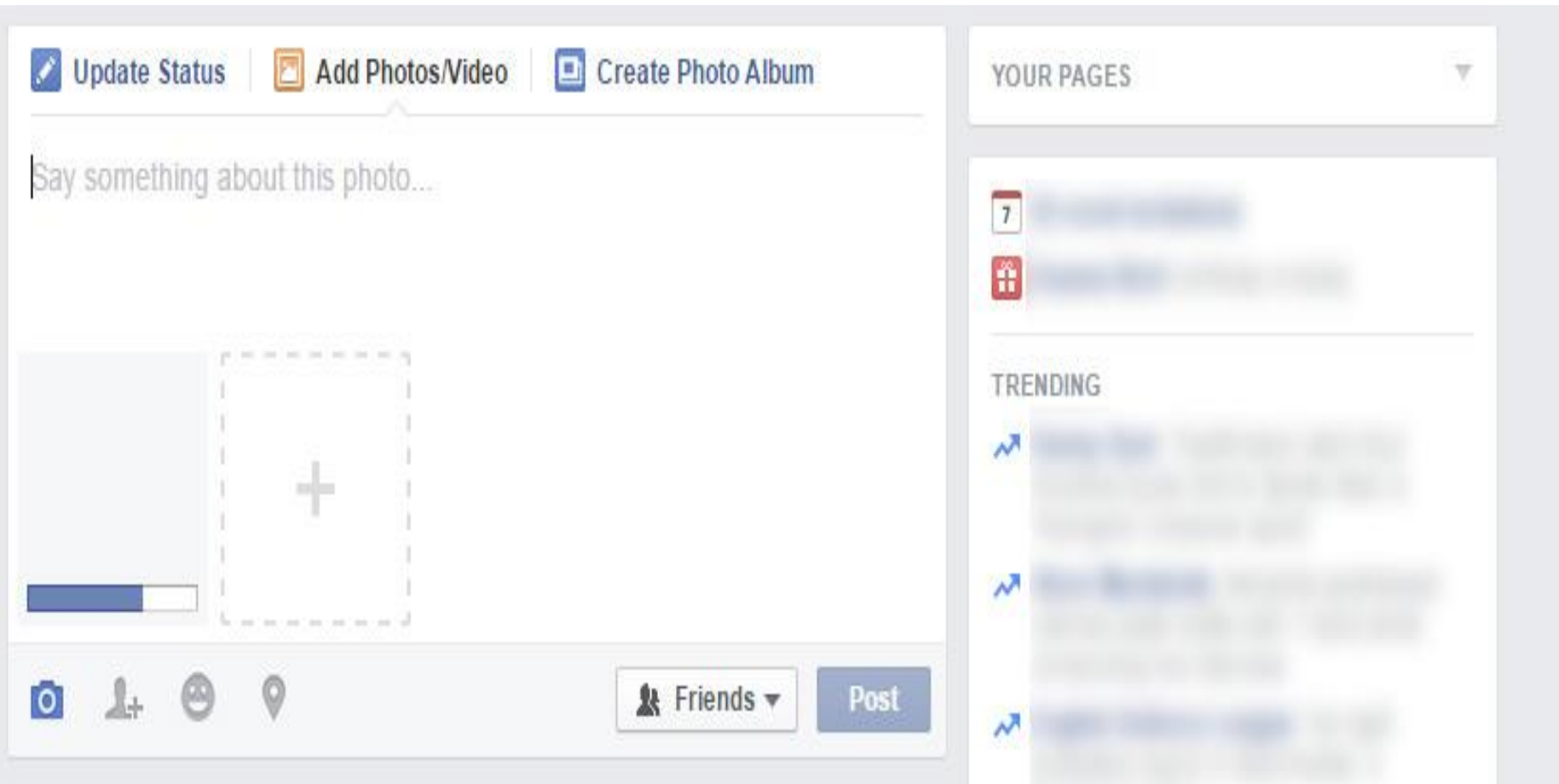
The image shows a screenshot of the Kickstarter website's project setup progress bar. The progress bar consists of seven steps: Basics, Rewards, Story, About you, Account, Preview, and Continue. The 'About you' step is currently selected and highlighted with a green checkmark. The 'Basics' step is also completed, indicated by a green checkmark. The 'Rewards', 'Story', 'Account', 'Preview', and 'Continue' steps are not yet completed, indicated by grey checkmarks. Below the progress bar, the text 'Start building your project!' is displayed, followed by the instruction 'Add an image, a goal, and other important details.'

KICKSTARTER Our Rules 

☒ Basics ☐ Rewards ☐ Story ☒ About you ☐ Account ☐ Preview ☐ Continue

Start building your project!
Add an image, a goal, and other important details.

1. Visibility of system status



1. Visibility of system status

The screenshot shows the Indiana University homepage. At the top, the university's name is in a large serif font. Below it is a navigation bar with links: Home, Prospective Students, Alumni & Donors, and Economic Development. A large red banner with the text 'We Are IU' is prominent. To the left of this banner is a photo of a family. To the right is a callout box stating 'preservation of top navigation is good.' Below the banner, there is a list of bullet points about the university's impact on Indiana. A second callout box points to the 'We Are IU' title, stating 'title shows current location, but no highlighting of the navigation to indicate directory'. On the left side of the page, there are several vertical menus: 'We are IU' (with links to Campuses, Medical Centers, Research), 'About IU' (with links to Admissions, Academics, Arts & Culture, Community Outreach, Emergency Preparedness, Employment, International Resources, Libraries, News & Information, Sports & Recreation, Technology), and 'Administration' (with links to President, Board of Trustees). At the bottom, there is a footer with the university's address, contact information, and copyright notice.

INDIANA UNIVERSITY

Home | Prospective Students | Alumni & Donors | Economic Development

We Are IU

preservation of top navigation is good.

title shows current location, but no highlighting of the navigation to indicate directory

- Nearly 258,000 IU alumni work in Indiana. More than 50 percent of Indiana's physicians, 40 percent of nurses, 64 percent of dentists are IU graduates.
- As part of the Indiana Life Sciences Initiative, IU is developing a business incubator and biomedical research center to take advantage of the state's economic prosperity and quality of life. With more than 10
- IU has recognized the central role of information technology in the 21st-century economy by creating the new School of Informatics.
- As part of Clarian Health Partners (University Hospital, Riley Hospital for Children, and Methodist hospital) and in collaboration with thousands of patients each year. In addition, many Hoosier citizens are treated at IU's eye care centers, dental clinics, and other health services.
- Through the Indiana Genomics Initiative (INGEN), IU scientists are using the genetic map published by the Human Genome Project to advance research in the life sciences.
- IU campuses across the state enrich the lives of Indiana residents with cultural offerings ranging from art exhibits, theatre, and music to lectures and performances.
- IU not only trains future teachers for Indiana's schools but also forms partnerships with schools and communities to improve education.

You can explore this Web site to find many more ways in which Indiana University benefits Hoosiers every day.

We are IU
Campuses
Medical Centers
Research

About IU
Admissions
Academics
Arts & Culture
Community Outreach
Emergency Preparedness
Employment
International Resources
Libraries
News & Information
Sports & Recreation
Technology

Administration
President
Board of Trustees

107 S. Indiana Ave., Bloomington, IN 47405-7000 | (317) 855-4848 | Comments | Disability Resources
Copyright © 2010 The Trustees of Indiana University | Copyright Complaints

1. Visibility of system status

The screenshot shows the Indiana University Emergency Preparedness website. At the top is the Indiana University logo and name. Below it is the title "Emergency Preparedness". The main content area features a large banner with the words "EXPLOSION", "TORNADO", "FIRE", "TERRORIST ATTACK", and "BIOHAZARD" in a grid. Below the banner is a yellow bar with the text "BE READY FOR ANYTHING" repeated four times. To the right of the banner is a box for "Influenza Information" with a link to "Learn more about Human Influenza A >>". Below the banner is a paragraph of text: "Indiana University wants you to be ready for anything. In the event of a disaster or emergency, this Web site will contain regularly updated news, instructions, and information. You will need to know what to do and what actions the university is taking in the following types of situations:". Below this text is a bulleted list of emergency types: Explosion, Severe Weather, Fire, Shooter, Terrorist Attack, and Biohazard. To the right of the list is a red box for "IU-Notify" with the text "Stay informed by managing your contact information through OneStart >>" and "Check out the IU-Notify FAQ to learn more about how IU-Notify works >>". Below the "IU-Notify" box is a section titled "Alert Status by Campus" with a list of campuses and their status: IU Bloomington (Normal), IUPUI (NORMAL), IPFW (Normal), IUPUC (Normal), IU East (Normal), and IU Kokomo (Normal). A callout box points to the top of the page, stating: "The navigation is lost completely on this page. Even though this page was accessed from the main page, we lose the main navigation bar, so there's no way to get back to where we came from."

Emergency Preparedness

EXPLOSION
TORNADO
FIRE
TERRORIST ATTACK
BIOHAZARD

BE READY FOR ANYTHING BE READY FOR ANYTHING BE READY FOR ANYTHING BE READY FOR ANYTHING

Indiana University wants you to be ready for anything. In the event of a disaster or emergency, this Web site will contain regularly updated news, instructions, and information. You will need to know what to do and what actions the university is taking in the following types of situations:

- Explosion
- Severe Weather
- Fire
- Shooter
- Terrorist Attack
- Biohazard

During critical situations, IU faculty, staff, and students will receive information and instructions directly through IU-Notify, an integrated e-mail.

Influenza Information
Learn more about Human Influenza A >>

IU-Notify
Stay informed by managing your contact information through OneStart >>
Check out the IU-Notify FAQ to learn more about how IU-Notify works >>

Alert Status by Campus

- IU Bloomington
Normal
- IUPUI
NORMAL
- IPFW
Normal
- IUPUC
Normal
- IU East
Normal
- IU Kokomo

The navigation is lost completely on this page. Even though this page was accessed from the main page, we lose the main navigation bar, so there's no way to get back to where we came from.

2. Match between system and the real world

- Speak the user's language using terms and concepts that are familiar to the intended audience. Information should be organized naturally and logically based on what users are accustomed to seeing in the real world.
- Does the system
 - speak users' language?
 - follow real world convention?

Example: When designing a website for children, use terms with which they are familiar and display information in formats they are used to seeing.

2. Match between system and the real world

The screenshot displays the Spotify Premium desktop application. The interface is dark-themed. On the left is a sidebar with navigation options: MAIN (Browse, Activity, Radio, Top Lists, Messages, Play Queue, Devices, App Finder, Lazify, Official Comedy, Pitchfork, Songkick Concerts) and YOUR MUSIC (Songs, Albums, Artists, Local Files, New Playlist, Starred). The main area is titled 'Play Queue' and shows a list of tracks playing from a playlist. The tracks are listed in a table with columns for TRACK, ARTIST, TIME, POPULARITY, and ALBUM. The first track is 'Expanding Anyway' by Morning Teleportat... with a duration of 3:42. The second track is 'Eyes The Same' by Morning Teleportat... with a duration of 3:13. The third track is 'In The Aeroplane Over The Sea' by Neutral Milk Hotel with a duration of 3:23. The fourth track is 'The King Of Carrot Flowers Pt. O...' by Neutral Milk Hotel with a duration of 2:00. The fifth track is 'Holland, 1945' by Neutral Milk Hotel with a duration of 3:13. The sixth track is 'Bloodbuzz Ohio' by The National with a duration of 4:36. The seventh track is 'We the Common (For Valerie Bol...' by Thao & The Get Do... with a duration of 3:37. The eighth track is 'Rox In The Box' by The Decemberists with a duration of 3:10. The ninth track is 'Don't Carry It All' by The Decemberists with a duration of 4:17. The tenth track is 'The World At Large' by Modest Mouse with a duration of 4:33. The eleventh track is 'The Good Times Are Killi...' by Modest Mouse with a duration of 4:17 and an 'EXPLICIT' label.

TRACK	ARTIST	TIME	POPULARITY	ALBUM
Expanding Anyway	Morning Teleportat...	3:42		Expanding Anyway
Eyes The Same	Morning Teleportat...	3:13		Expanding Anyway
In The Aeroplane Over The Sea	Neutral Milk Hotel	3:23		In The Aeroplane Ov...
The King Of Carrot Flowers Pt. O...	Neutral Milk Hotel	2:00		In The Aeroplane Ov...
Holland, 1945	Neutral Milk Hotel	3:13		In The Aeroplane Ov...
Bloodbuzz Ohio	The National	4:36		High Violet (Expande...
We the Common (For Valerie Bol...	Thao & The Get Do...	3:37		We the Common
Rox In The Box	The Decemberists	3:10		The King Is Dead
Don't Carry It All	The Decemberists	4:17		The King Is Dead
The World At Large	Modest Mouse	4:33		Good News For Peop...
The Good Times Are Killi... [EXPLICIT]	Modest Mouse	4:17		Good News For Peop...

2. Match between system and the real world

Prospective Students

Each year, Indiana University welcomes thousands of new students to our eight campuses. Most are first-year college students, but the quality of life is enhanced by transfer students, international students, and returning students who also join our ranks every year. Discover all that IU offers for undergraduate students.

IU Bloomington

Office of Admissions
300 North Jordan Ave.
Bloomington, IN 47405-1106
(812) 855-0661
E-mail: iuadmit@indiana.edu

- Campus Profile
- Bursar (fees/costs)
- Student Financial Assistance
- Campus Map and Building List

why use the word "bursar" when you need to clarify with (fees/costs)?

- Photo Tour
- Registrar

IUPUI Indianapolis

Enrollment Services
425 University Blvd.
Cavanaugh Hall 129
Indianapolis, IN 46202-5143
(317) 274-4591
E-mail: apply@iupui.edu

- Campus Profile
- Bursar (fees/costs)
- Campus Maps and Directions
- Orientation
- Photo Tour
- Registrar
- Student Financial Aid Services

IU East

Office of Admissions
2225 Cheater Blvd

- Campus Profile
- Academic Advising
- Bursar (fees/costs)
- Campus Maps and Directions

2. Match between system and the real world

The screenshot shows the Indiana University website with a navigation menu on the left and a main content area on the right. The navigation menu lists various categories, and the main content area shows the 'We are IU' section with links to different campuses. A callout box points to the 'Research' link in the navigation menu, which leads to a page titled 'what is IUPUI?'. Another callout box points to the 'Prospective Students' link in the navigation menu, which leads to a page titled 'extreme variation of language used to describe the different campuses may confuse users'.

Navigation Menu (Left):

- Campuses ▶
- Medical Centers ▶
- Research ▶
- About IU ▶
- Admissions ▶
- Academics ▶
- Arts & Culture ▶
- Community Outreach ▶
- Emergency Preparedness ▶
- Academics ▶
- Arts & Culture ▶
- Community Outreach ▶
- Emergency Preparedness ▶
- Library Resources ▶
- Library Resources ▶
- Libraries ▶
- News & Information ▶
- Sports & Recreation ▶
- Technology ▶
- Administration ▶
- President ▶
- Board of Trustees ▶

Main Content Area (Right):

INDIANA UN

Prospective Students | Alumni & Donors

We are IU

- Campuses ▶
- Medical Centers ▶
- Research ▶
- About IU ▶
- Admissions ▶
- Academics ▶
- Arts & Culture ▶
- Community Outreach ▶
- Emergency Preparedness ▶
- Diversity Resources ▶
- International Resources ▶
- Libraries ▶
- News & Information ▶
- Sports & Recreation ▶
- Technology ▶

Callout Boxes:

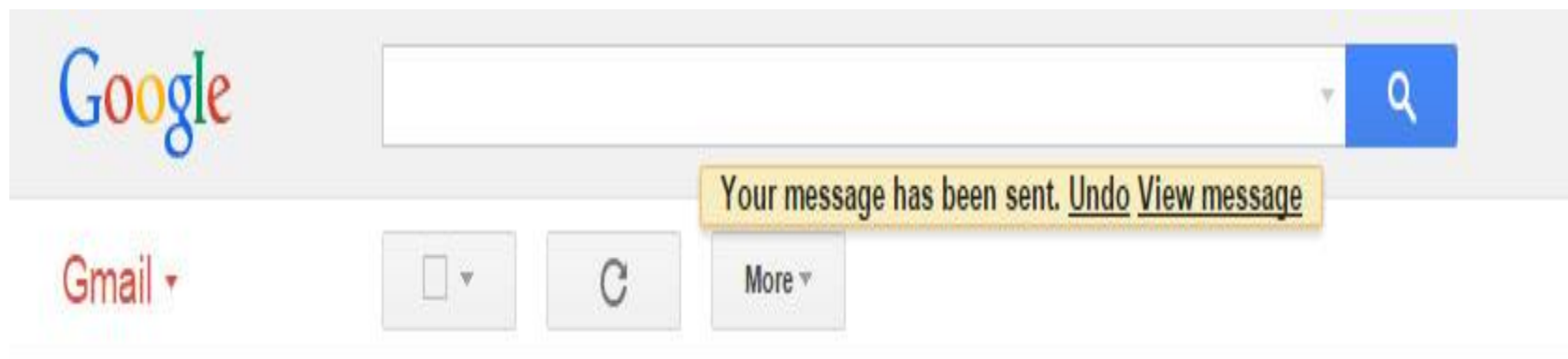
- what is IUPUI?** (points to Research link)
- extreme variation of language used to describe the different campuses may confuse users** (points to Prospective Students link)

3. User Control and Freedom

- Users should experience perceived control as they interact with the system. Users should have the ability to “go back” and fix mistakes without hassle.
- *Does the system*
 - *provide clearly marked exists?*
 - *support undo and redo*

Example: Provide the functionality to Undo and Redo actions and to easily exit the system.

3. User Control and Freedom



3. User Control and Freedom

Comments

Your information needs and comments about this site are important to us. If you have a question is directed to the appropriate office or department.

Select a topic:

--- Choose an option ---
--- Choose an option ---
Give us feedback on the search function
Question / Comment about the IU Gateway
Add your site to the IU Gateway or update your existing information
Question about admissions, scholarships, or how to apply
Comment about Indiana University
Question / comment about a news item

restrictive use of topics on the comments page.

404 error if you make an error in your first submission

Your name:

Your e-mail address

Submit

Error

Error 404 - Page not found

Sorry for the inconvenience, the page you requested could not be found.

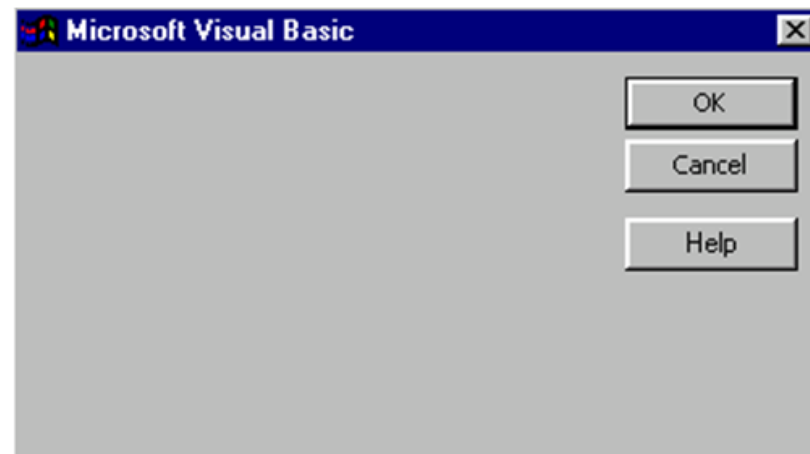
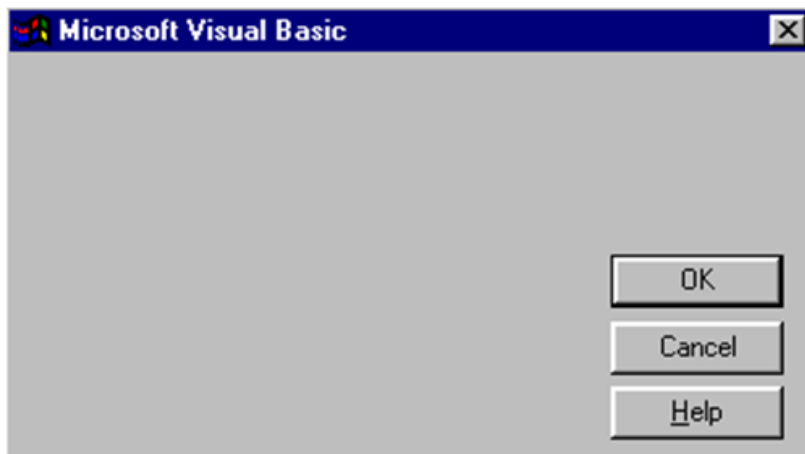
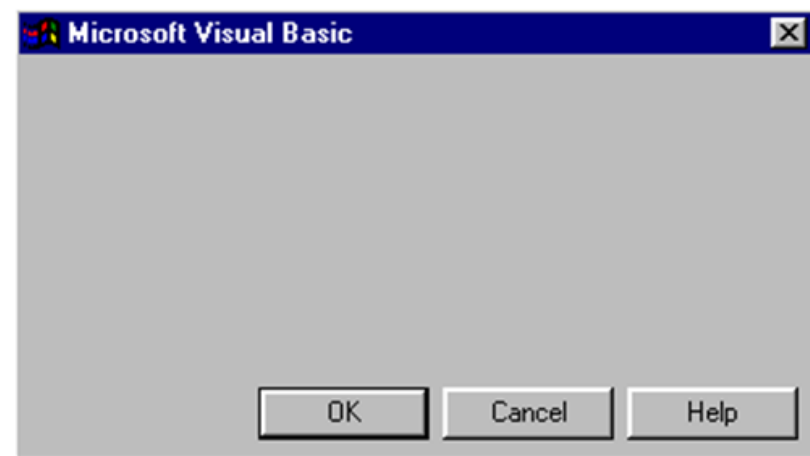
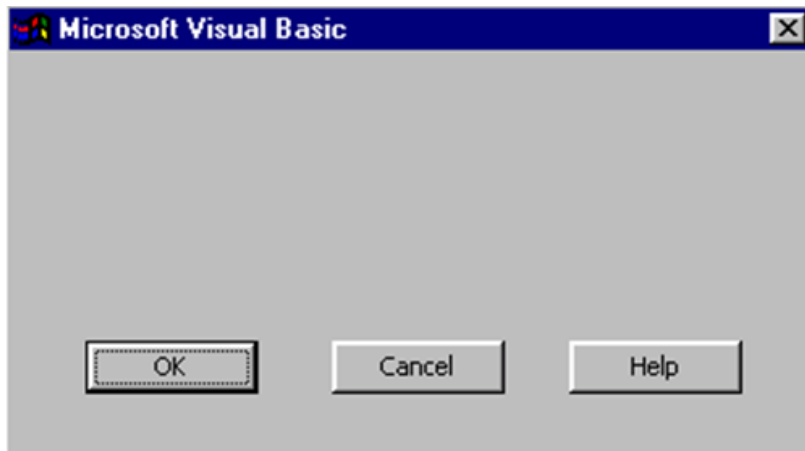
If the site you are in offers a site map or 'A-Z Big List' try those first. Otherwise, try conducting

4. Consistency and Standards

- User controls, icons, terminology, and error messaging should be consistent throughout the interface. Where appropriate, industry and platform standards should be applied.
- Does the system use the same term to refer to the same thing?
- Does the system let users perform the same action by the same UI?

Example: Use icons with which people are familiar, rather than creating new designs that mean the same thing. Users are more likely to be scanning around the page for the phrase “About Us” than “Get to know the team”.

4. Consistency and Standards



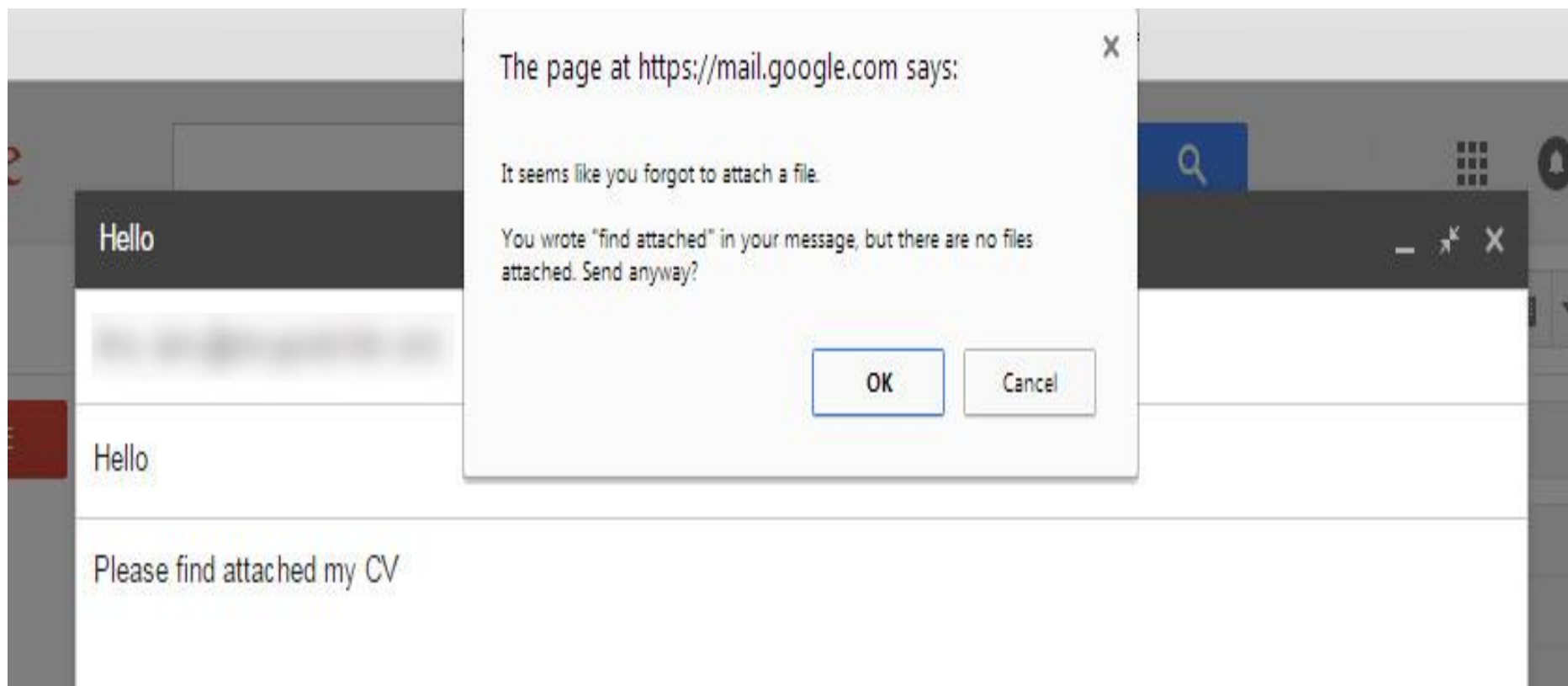
5. Error Prevention

- Prevent user errors by identify problem areas for typical users and re-designing it to more clearly communicate the consequences of users' actions.
- When deleting information that may be difficult to recreate, offer users a confirmation to delete the data. Additionally, provide the ability to Undo actions that users could accidentally commit and, consequently, lose important information.
- Does the system eliminate or check for error-prone conditions?

Example: does an email client help users avoid common errors such as

- typing the wrong address
- making spelling mistakes
- missing an attachment

5. Error Prevention

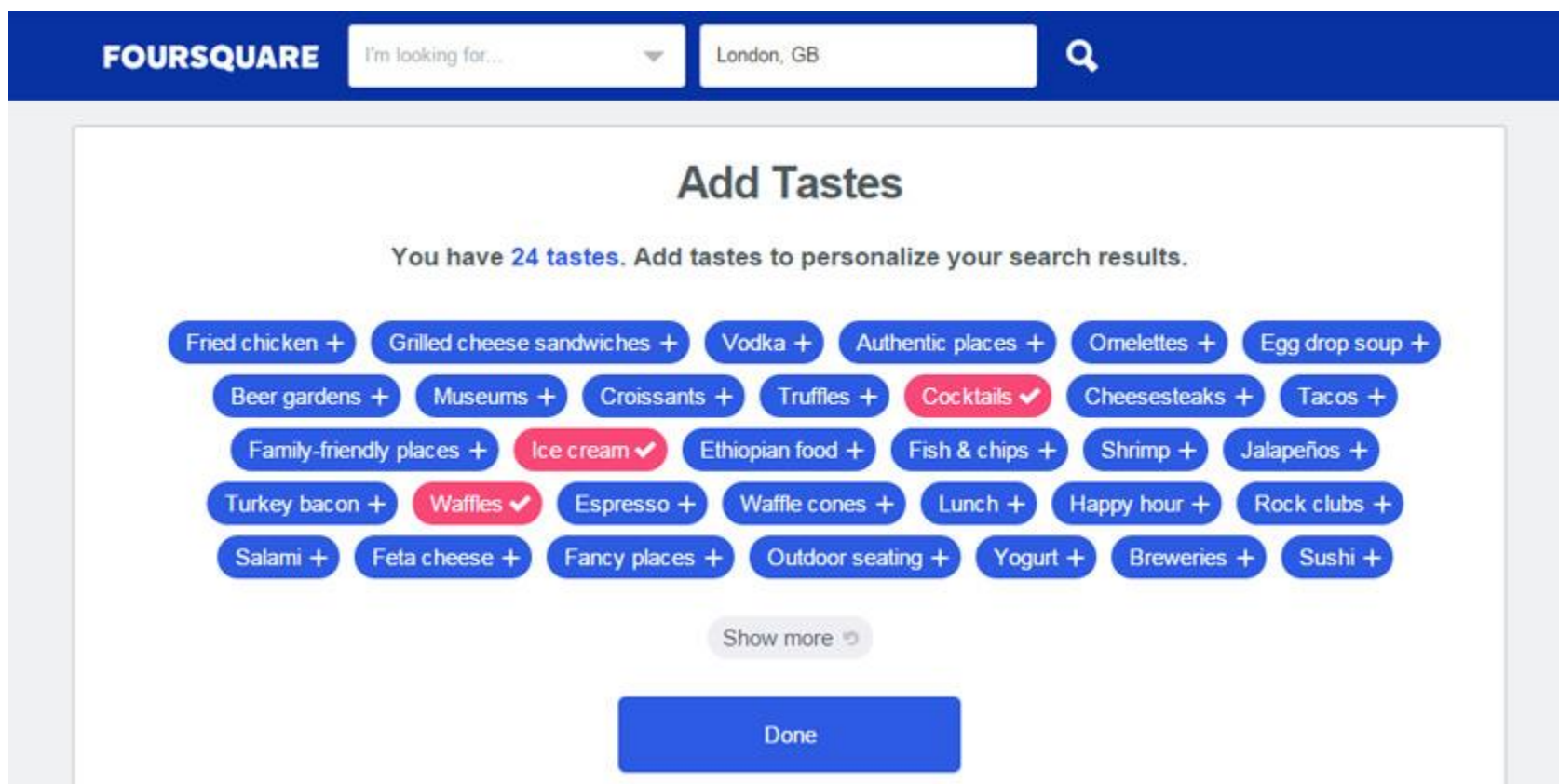


6. Recognition rather than recall

- Reduce the memory load of users by presenting familiar icons, actions, and options whenever possible.
- Do not require the user to recall information from one screen to another.
- Use mouse-over tooltips to describe the functionality of icons which may be unfamiliar.
- Does the system present options for users to recognize the one they want?

Example: On a web form, allow easy access to previously entered information, such as serial numbers, so the user does not need to recall the information or write it down.

6. Recognition rather than recall

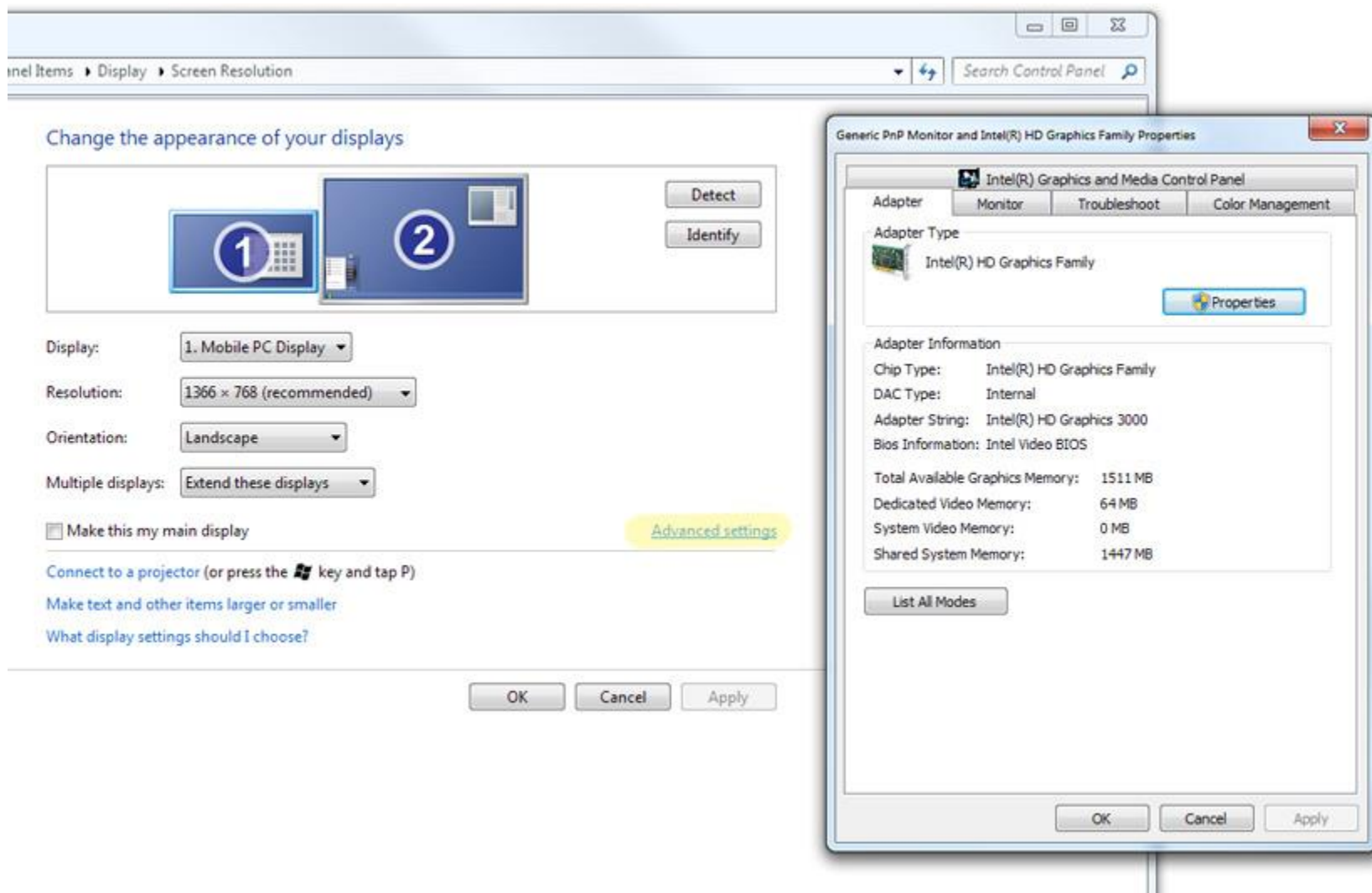


7. Flexibility and efficiency of use

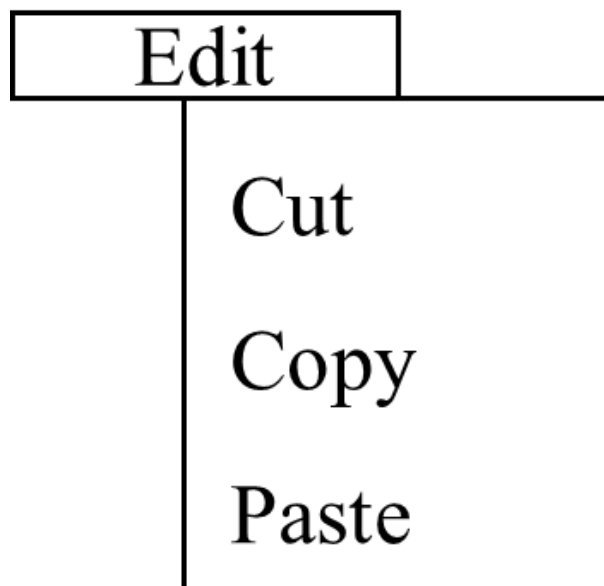
- Novice and expert users use systems differently. The system should be easy and efficient to use by novices and experts alike.
- Provide “accelerators” for expert users to more efficiently navigate your application to complete the most frequent tasks.
- Does the system provide ways for users to access a file or application quickly?
- Does the system provide shortcuts users can learn to do things faster?

Example: An accelerator can be a keystroke shortcut, such as Macintosh’s Command+Q to quit an application.

7. Flexibility and efficiency of use



7. Flexibility and efficiency of use



KB Shortcut:

Ctrl + X

Ctrl + C

Ctrl + V

8. Aesthetic and minimalist design

- Avoid displaying excessive information and design elements, as they will visually compete with more relevant information on the screen.
- Does the system minimize irrelevant or rarely needed elements?
- Does the system look aesthetically appealing?

Example: Background graphics can make viewing text difficult.

8. Aesthetic and minimalist design



What advice would you give your younger self? [#DearMe](#)

8. Aesthetic and minimalist design

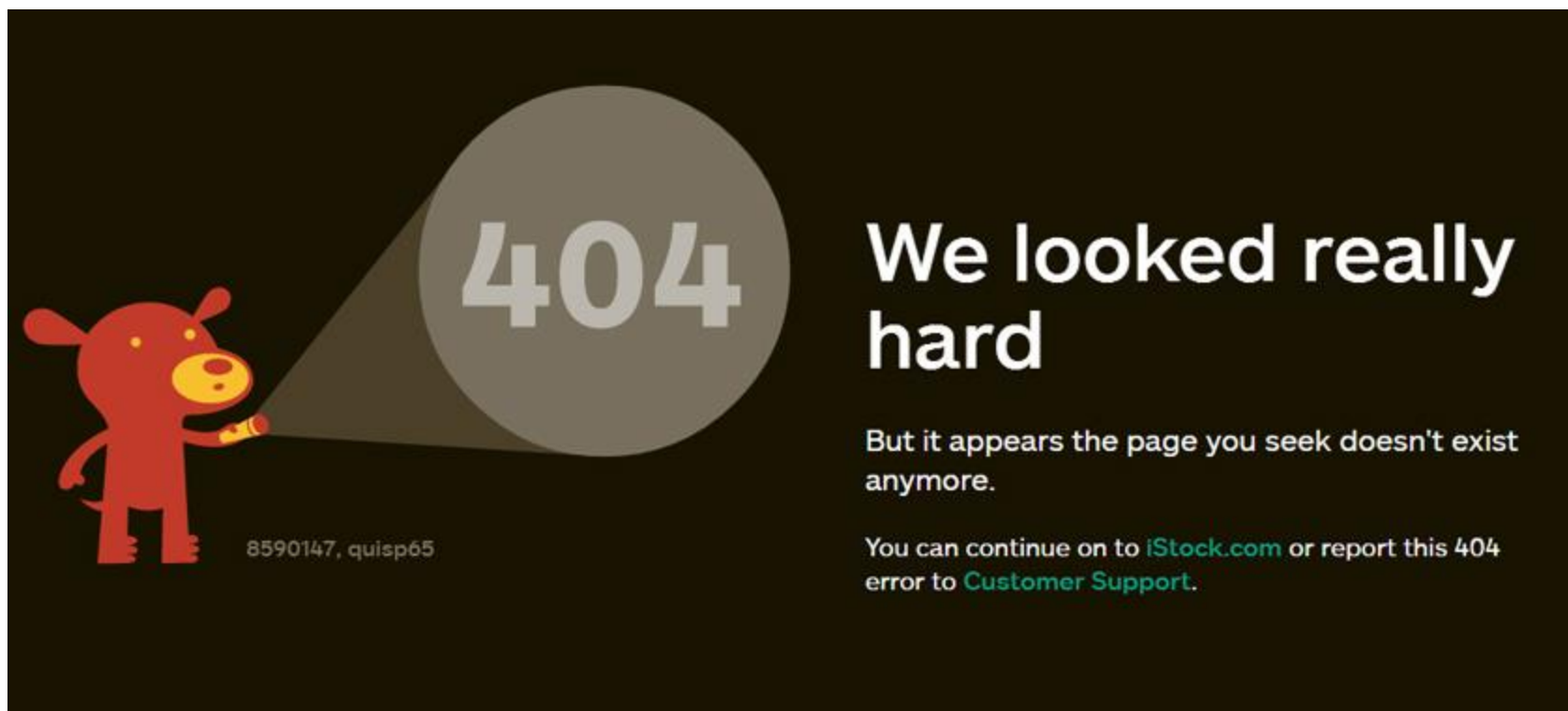
Form Title -- (appears above URL in most browsers and is used by WWW search)		Background Color:
Q&D Software Development Order Desk		FFFBF0
Form Heading -- (appears at top of Web page in bold type)		Text Color:
Q&D Software Development Order Desk ☒ Center		000080
E-Mail responses to (will not appear on)	Alternate (for mailto forms only)	Background Graphic
dversch@q-d.com		
Text to appear in Submit button	Text to appear in Reset button	☐ Mailto
Send Order	Clear Form	☒ CGI
Scrolling Status Bar Message (max length = 200 characters)		
WebMania 1.5b with Image Map Wizard is here!		
<< Prev Tab		Next Tab >>

9. Help Users Recognize, Diagnose, and Recover from Errors

- Present error messages that give users instructions about how to recover from an error, rather than cryptic codes that users do not understand.
- Does the system explain well why an error occurs and how to fix it?

Example: If the user enters an invalid email address on a web form that requests the address, the error message could read, "That email address is not in our records. Please enter an email address in this format: email@address.com."

9. Help Users Recognize, Diagnose, and Recover from Errors



10. Help and Documentation

- It is usually best to design an interface to be so simple to use that Help and Documentation is unnecessary; however, there may be times when it is necessary to access Help.
- In those cases, Help documentation and user support should be easy to search, and instructions should be easy to follow.
- It may be useful to provide video tutorials and examples for complex procedures or controls that are hard to locate.
- Does the system help? Does the system provide help where and when it is needed? Is the system adequately documented?

Example: If there is not enough reason to produce an entire Help section, and there are a couple form fields that may be confusing to some users, it is appropriate to include “in-line help” in the form of a link that opens a small help dialogue next to the form field. Alternatively, next to a form field text box or field label, provide users an example of how to input the information using the required formatting, such as entering a phone number as xxx-xxx-xxxx.

10. Help and Documentation

Etsy

Search for items or shops

Search

Browse ▾

Register

Sign In



How can we help you?

Search for help by topics, keywords or phrases

Search help

Getting Started

[Etsy Guidelines](#)

[Confirming Your Etsy Account](#)

[Search for Items and Shops](#)

[Purchasing an Item](#)

[Contacting a Seller](#)

Tools and Features

[Receiving Gift Cards](#)

[Reporting a Site Bug](#)

[Get Faster Shipping](#)

[Refunds and Returns](#)

[Report a Problem with an Order](#)

Become a Seller

[Setting up a Shop](#)

[Getting Paid](#)

[Fees for Selling on Etsy](#)

[Listing a Physical Item](#)

[Choosing Payment Methods in Your Shop](#)



Contacting the Seller

Got a question about an item? We'll show you how to ask the seller.



Sign Up to Sell

Turn your passion into a business, open up an Etsy shop.

Still have questions?

Check the FAQs

Find an overview of popular help topics.

Ask in the Forums

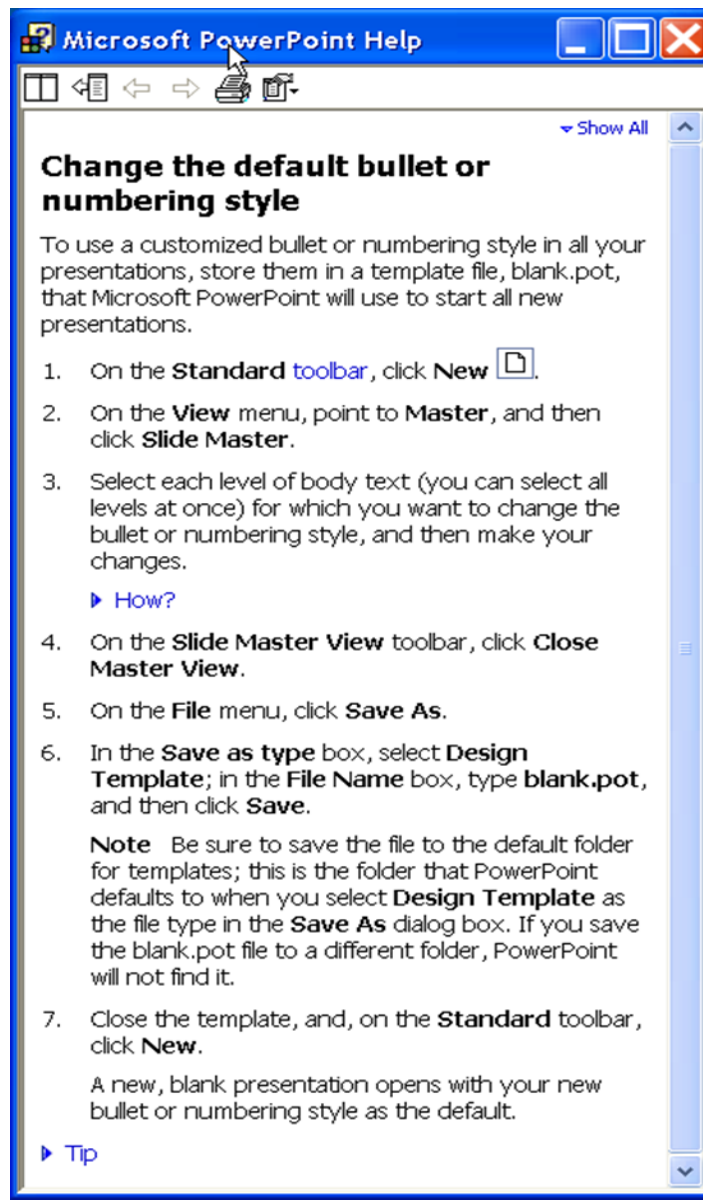
Learn from our knowledgeable community.

Site Policies

Read the ins and outs of selling on Etsy.

No luck? Contact us and we'll get back to you as soon as possible.

10. Help and Documentation



Heuristic evaluation

10 Usability Heuristics Video:

<https://www.youtube.com/watch?v=hWc0Fd2AS3s>

Further Reading Materials (Must-Read)

- Nielsen's papers on his website:
<http://www.useit.com/papers/heuristic/>
- 1 document (given during the lecture slides)
- <http://designmodo.com/usability-heuristic-evaluation/>
- http://www.mattsoave.com/old/cogs187a/iu_site_eval/1visibility.html
- Search in Google and read articles/blog/documents related to heuristic evaluation and its example cases.



Evaluating through user Participation

Evaluating through User Participation

1. Laboratory studies
2. Field studies

Laboratory studies

- Advantages:
 - specialist equipment available
 - uninterrupted environment
- Disadvantages:
 - lack of context
 - difficult to observe several users cooperating
- Appropriate
 - If system location is dangerous or impractical (space station)
 - For constrained single-user tasks (systems)
 - To allow controlled manipulation of use
 - To compare alternative designs within a controlled context

Field Studies

- Advantages:
 - natural environment
 - context retained (though observation may alter it)
 - longitudinal studies possible
- Disadvantages:
 - distractions
 - noise
- Appropriate
 - where context is crucial for longitudinal studies

Evaluating Implementations

Requires an artefact:

- simulation
- prototype
- full implementation

Experimental Evaluation

- Controlled evaluation of specific aspects of interactive behaviour
- Evaluator chooses hypothesis to be tested
- A number of experimental conditions are considered which differ only in the value of some controlled variable.
- Changes in behavioural measure are attributed to different conditions

Experimental factors

- Subjects/participants
 - who – representative, sufficient sample
- Variables
 - things to modify and measure
- Hypothesis
 - what you'd like to show
- Experimental design
 - how you are going to do it

Participants

- Choice of participants is vital to the success of any experiment
- Should be chosen to match the expected user population as closely as possible
- Computer science UG students are not a good choice to test an application developed for general public.
- Sample size must be large enough to be representative of population, taking into account the design of the experiment and the statistical methods chosen.

Variables

- Independent Variable (IV)
characteristic changed to produce different conditions
e.g. interface style, number of menu items
- Dependent Variable (DV)
characteristics measured in the experiment
e.g. time taken, number of errors.

Hypothesis

- prediction of outcome
 - framed in terms of IV and DV

e.g. “error rate will increase as font size decreases”
- null hypothesis:
 - states no difference between conditions
 - aim is to disprove this

e.g. null hyp. = “no change with font size”

Experimental design

- within groups design
 - each subject performs experiment under each condition.
 - transfer of learning possible
 - less costly and less likely to suffer from user variation.
- between groups design
 - each subject performs under only one condition
 - no transfer of learning
 - more users required
 - variation can bias results.

Analysis of data

- Before you start to do any statistics:
 - look at data
 - save original data
- Choice of statistical technique depends on
 - type of data
 - information required
- Type of data
 - discrete - finite number of values
 - continuous - any value

Analysis - types of test

- parametric
 - assume normal distribution
 - robust
 - powerful
- non-parametric
 - do not assume normal distribution
 - less powerful
 - more reliable
- contingency table
 - classify data by discrete attributes
 - count number of data items in each group

Analysis of data (cont.)

- What information is required?
 - is there a difference?
 - how big is the difference?
 - how accurate is the estimate?
- Parametric and non-parametric tests mainly address first of these

Experimental studies on groups

More difficult than single-user experiments

Problems with:

- subject groups
- choice of task
- data gathering
- analysis

Subject groups

larger number of subjects

⇒ more expensive

longer time to `settle down`

... even more variation!

difficult to timetable

so ... often only three or four groups

The task

must encourage cooperation

perhaps involve multiple channels

options:

- creative task e.g. *'write a short report on ...'*
- decision games e.g. desert survival task
- control task e.g. ARKola bottling plant

Data gathering

several video cameras
+ direct logging of application

problems:

- synchronisation
- sheer volume!

one solution:

- record from each perspective

Analysis

N.B. vast variation between groups

solutions:

- within groups experiments
- micro-analysis (e.g., gaps in speech)
- anecdotal and qualitative analysis

look at interactions between group and media

controlled experiments may 'waste' resources!

Field studies

Experiments dominated by group formation

Field studies more realistic:

distributed cognition $\Rightarrow$ work studied in context

real action is *situated action*

physical and social environment both crucial

Contrast:

psychology – controlled experiment

sociology and anthropology – open study and rich data

Observational Methods

Think Aloud

Cooperative evaluation

Protocol analysis

Automated analysis

Post-task walkthroughs

Think Aloud

- user observed performing task
- user asked to describe what he is doing and why, what he thinks is happening etc.
- Advantages
 - simplicity - requires little expertise
 - can provide useful insight
 - can show how system is actually use
- Disadvantages
 - subjective
 - selective
 - act of describing may alter task performance

Cooperative evaluation

- variation on think aloud
- user collaborates in evaluation
- both user and evaluator can ask each other questions throughout
- Additional advantages
 - less constrained and easier to use
 - user is encouraged to criticize system
 - clarification possible

Protocol analysis

Methods for recording user actions include the following:

- paper and pencil – cheap, limited to writing speed
- audio – good for think aloud, difficult to match with other protocols
- video – accurate and realistic, needs special equipment, obtrusive
- computer logging – automatic and unobtrusive, large amounts of data difficult to analyze
- user notebooks – coarse and subjective, useful insights, good for longitudinal studies
- Mixed use in practice.
- audio/video transcription difficult and requires skill.
- Some automatic support tools available

automated analysis - Experimental Video Annotator (EVA)

- Workplace project
- Post task walkthrough
 - user reacts on action after the event
 - used to fill in intention
- Advantages
 - analyst has time to focus on relevant incidents
 - avoid excessive interruption of task
- Disadvantages
 - lack of freshness
 - may be post-hoc interpretation of events

post-task walkthroughs

- transcript played back to participant for comment
 - immediately → fresh in mind
 - delayed → evaluator has time to identify questions
- useful to identify reasons for actions and alternatives considered
- necessary in cases where think aloud is not possible

Query Techniques

Interviews
Questionnaires

Interviews

- analyst questions user on one-to-one basis usually based on prepared questions
- informal, subjective and relatively cheap
- Advantages
 - can be varied to suit context
 - issues can be explored more fully
 - can elicit user views and identify unanticipated problems
- Disadvantages
 - very subjective
 - time consuming

Questionnaires

- Set of fixed questions given to users
- Advantages
 - quick and reaches large user group
 - can be analyzed more rigorously
- Disadvantages
 - less flexible
 - less probing

Questionnaires (ctd)

- Need careful design
 - what information is required?
 - how are answers to be analyzed?
- Styles of question
 - general
 - open-ended
 - scalar
 - multi-choice
 - ranked

Physiological methods

Eye tracking

Physiological measurement

eye tracking

- head or desk mounted equipment tracks the position of the eye
- eye movement reflects the amount of cognitive processing a display requires
- measurements include
 - fixations: eye maintains stable position. Number and duration indicate level of difficulty with display
 - saccades: rapid eye movement from one point of interest to another
 - scan paths: moving straight to a target with a short fixation at the target is optimal

physiological measurements

- emotional response linked to physical changes
- these may help determine a user's reaction to an interface
- measurements include:
 - heart activity, including blood pressure, volume and pulse.
 - activity of sweat glands: Galvanic Skin Response (GSR)
 - electrical activity in muscle: electromyogram (EMG)
 - electrical activity in brain: electroencephalogram (EEG)
- some difficulty in interpreting these physiological responses - more research needed

Choosing an Evaluation Method

when in process:	design vs. implementation
style of evaluation:	laboratory vs. field
how objective:	subjective vs. objective
type of measures:	qualitative vs. quantitative
level of information:	high level vs. low level
level of interference:	obtrusive vs. unobtrusive
resources available:	time, subjects, equipment, expertise